LIST OF FORMULAE

Volume of a prism — $V = Ah$ where A is the area of the cross-section and h is the perpendicular length.

Volume of a cylinder — $V = \pi r^2 h$ where r is the radius of the base and h is the perpendicular height.

Volume of a right pyramid — $V = \frac{1}{3}Ah$ where A is the area of the base and h is the perpendicular height.

Circumference — $C = 2\pi r$ where r is the radius of the circle.

Arc length — $S = \frac{\theta}{360} \times 2\pi r$ where θ is the angle subtended by the arc, measured in degrees.

Area of a circle — $A = \pi r^2$ where r is the radius of the circle.

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Area of a sector — $A = \frac{\theta}{360} \times \pi r^2$ where θ is the angle of the sector, measured in degrees.

Area of a trapezium — $A = \frac{1}{2}(a + b)\,h$ where a and b are the lengths of the parallel sides and h is the perpendicular distance between the parallel sides.

Roots of quadratic equations — If $ax^2 + bx + c = 0$,

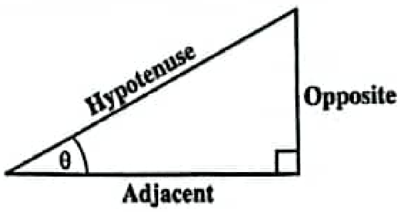
then $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Trigonometric ratios

$\sin\theta = \dfrac{\text{length of opposite side}}{\text{length of hypotenuse}}$

$\cos\theta = \dfrac{\text{length of adjacent side}}{\text{length of hypotenuse}}$

$\tan\theta = \dfrac{\text{length of opposite side}}{\text{length of adjacent side}}$



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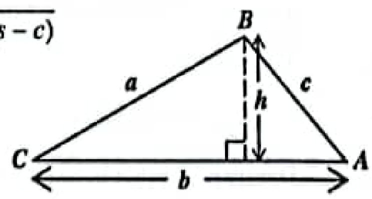
Area of a triangle — Area of $\Delta = \frac{1}{2}bh$ where b is the length of the base and h is the perpendicular height.

Area of $\Delta ABC = \frac{1}{2}ab\sin C$

Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

where $s = \dfrac{a+b+c}{2}$



Sine rule — $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine rule — $a^2 = b^2 + c^2 - 2bc\cos A$

1. What percentage of 40 is 8?

(A) 5%
(B) 20%
(C) 32%
(D) 150%

2. The value of the digit 2 in the number 748.621 is

(A) $\frac{2}{10^2}$

(B) $\frac{2}{10^3}$

(C) 2×10^{-1}

(D) $2\,000 \times 10^{-3}$

3. $\sqrt{17^2 - 15^2}$ is equal to

(A) 1
(B) 2
(C) 8
(D) 16

4. In a school, the ratio of the number of pupils to the number of teachers is 20:1. If the number of pupils is 840, how many teachers are there?

(A) 40
(B) 42
(C) 820
(D) 840

5. A bag of apples can be shared equally among either 6, 10 or 15 children. The MINIMUM number of apples that is likely to be in the bag is

(A) 30
(B) 31
(C) 60
(D) 90

6. 99×101 has the same value as

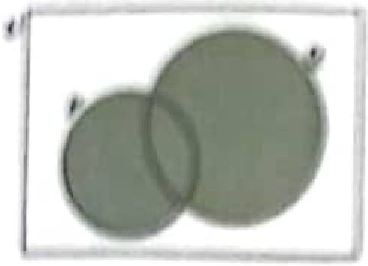
(A) $(99 \times 100) + 1$
(B) $(99 \times 100)\,(99 \times 1)$
(C) $(99 \times 100) - (99 \times 1)$
(D) $(99 \times 100) + (99 \times 1)$

7. Which of the following sets is defined by

$$\{x \in Z : -2 \leq x \leq 4\}?$$

(A) $\{1, 2, 3, 4\}$
(B) $\{0, 1, 2, 3, 4\}$
(C) $\{-1, 0, 1, 2, 3\}$
(D) $\{-2, -1, 0, 1, 2, 3, 4\}$

Item 8 refers to the following Venn diagram which shows 2 intersecting sets, P and Q. In the Venn diagram, $n(P) = 6$, $n(Q) = 8$ and $n(P \cap Q) =$ [illegible]



8. The number of elements in the shaded region of the Venn diagram is

(A) 9
(B) 10
(C) 14
(D) 18

9. In which of the following pairs of sets is one set NOT a subset of the other?

(A) P = {multiples of 12} and
Q = {factors of 3}

(B) E = {factors of 6} and
F = {factors of 12}

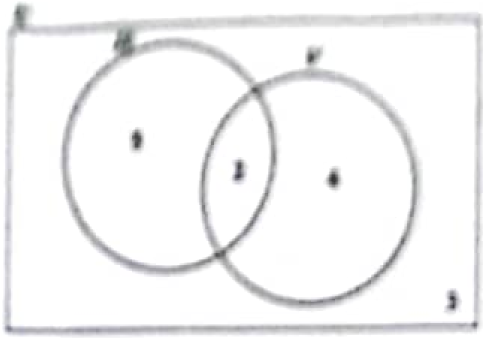
(C) X = {whole numbers} and
Y = {rational numbers}

(D) G = {multiples of five} and
H = {multiples of ten}

10. All students in a class play Scrabble or Checkers or both. Sixteen per cent of the students play BOTH Scrabble and Checkers. The number of students who play Checkers only is TWICE the number of students who play Scrabble only. What percentage of students play Checkers but NOT Scrabble?

(A) 28
(B) [illegible]
(C) 56
(D) 72

Item 11 refers to the following Venn diagram which shows 2 intersecting sets. The number of students in each set is indicated.



11. In the Venn diagram,

U = {students who play games}
H = {students who play hockey}
V = {students who play volleyball}.

How many students play EITHER hockey OR volleyball but not both?

(A) 6
(B) 7
(C) 9
(D) 11

12. The set of positive integers that are divisible by 7 is an example of

(A) an improper set
(B) an infinite set
(C) an empty set
(D) a finite set

Items 13 and 14 refer to the following diagram which shows part of a shopping bill.

Item	Unit Cost Price	Total Cost Price
4 kg flour	*P*	\$12.40
2 kg sugar	\$3.60	\$ 7.20
8 kg rice	*Q*	*T*
Subtotal		\$40.40
VAT (8%)		xxxxxx
Total		xxxxxx

13. The correct values of *P*, *Q* and *T* are

	P	*Q*	*T*
(A)	\$0.99	\$1.66	\$19.60
(B)	\$3.10	\$6.70	\$20.80
(C)	\$4.05	\$5.50	\$19.60
(D)	\$3.10	\$2.60	\$20.80

14. According to the bill, the amount paid for VAT is

(A) \$ 3.10
(B) \$ 3.23
(C) \$37.17
(D) \$43.63

15. At a bank, EC\$2.60 is equivalent to US\$1.00. For every US\$1.00 exchanged, EC\$0.10 is deducted as an exchange tax. How many EC dollars will Leon receive if he exchanges US\$1 000.00?

(A) \$ 900.90
(B) \$2 360.34
(C) \$2 500.00
(D) \$2 600.00

16. A calculator which is marked at \$120 is sold for cash at a 30% discount. How much change would Susan receive if she pays for the calculator with a \$100 bill?

(A) \$16
(B) \$20
(C) \$28
(D) \$36

17. By selling a bag for \$1 140, Vishal incurred a loss of 5%. At what price should he have sold the bag to gain a profit of 5%?

(A) \$1 197.00
(B) \$1 254.00
(C) \$1 256.85
(D) \$1 260.00

18. The cost of a machine is estimated to be decreasing at the rate of 10% every year. If it currently costs \$6 000, what will be the estimated value of the machine after 2 years?

(A) \$1 140
(B) \$4 800
(C) \$4 860
(D) \$5 400

19. A man pays 60 cents for every 200 m^3 of gas used, plus a fixed charge. If he pays \$178.75 when he uses 55 000 m^3 of gas, how much is the fixed charge?

(A) \$ 13.75
(B) \$ 14.35
(C) \$151.25
(D) \$165.00

20. The compound interest on \$12 000 for 2 years at 10% per annum, compounded annually, is

(A) \$ 1 320
(B) \$ 2 520
(C) \$13 200
(D) \$14 520

21. The product of a number p and its reciprocal may be written as

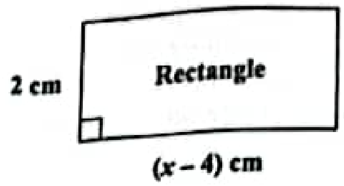
(A) $p^2 \times (-p)$

(B) $p \times (-p)$

(C) $p^2 \times \frac{1}{p}$

(D) $p \times \frac{1}{p}$

Item 22 refers to the following diagram of a rectangle.



22. The area of the rectangle, in cm^2, is x^2. The equation that may be used to find the value of x is

(A) $x^2 = 2(x - 4)$
(B) $x^2 = (x - 2)(x - 4)$
(C) $x^2 = (x - 4)(x + 2)$
(D) $x^2 = 2(x - 4)(x - 2)$

23. Althea, Bob and Chris collect shells. Althea has p shells, Bob has TWICE as many shells as Althea and Chris has 4 more than the total number of shells that both Althea and Bob collected. Altogether, the total number of shells they have is

(A) $6p$
(B) $3p + 4$
(C) $6(p + 2)$
(D) $2(3p + 2)$

24. If $3 + \frac{2}{x} = 1$, then the value of x is

(A) -1

(B) $\frac{1}{5}$

(C) $\frac{1}{2}$

(D) 5

25. The value of $5^{n+1} \times 5^{n+2}$ when $n = -1$ is

(A) 1
(B) 5
(C) 10
(D) 25

26. If 40 students will take 20 days to paint a wall, how many days will it take to paint the wall if 10 more students are added, working at the same rate?

(A) 12
(B) 15
(C) 16
(D) 18

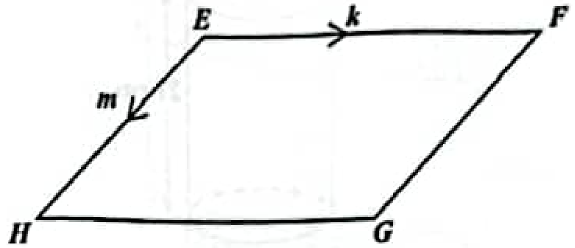
27. If $P = \begin{pmatrix} 2 & 5 \\ -1 & 3 \end{pmatrix}$, then the value of $|P|$ is

(A) 0
(B) 1
(C) 4
(D) 11

28. If $A = \begin{pmatrix} 1 & 2 & 5 & 4 \\ 6 & 1 & 3 & 7 \\ -2 & 3 & 2 & 9 \end{pmatrix}$, then the order of A is

(A) 2×3
(B) 3×2
(C) 3×4
(D) 4×3

Item 29 refers to the following diagram of a parallelogram, in which EF is parallel to HG, EH is parallel to FG, $\overrightarrow{EF} = k$ and $\overrightarrow{EH} = m$.



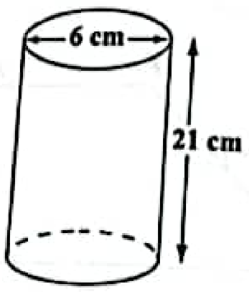
29. $\overrightarrow{EG}$ expressed in terms of k and m is

(A) $-m - k$
(B) $m - k$
(C) $k - m$
(D) $k + m$

30. If $5\begin{pmatrix} x \\ y \end{pmatrix} = 4\begin{pmatrix} 10 \\ 20 \end{pmatrix}$, then the values of x and y are

(A) $x = 4,\ y = 5$
(B) $x = 8,\ y = 16$
(C) $x = 2.5,\ y = 4$
(D) $x = 10,\ y = 20$

Item 31 refers to the following diagram which shows a cylinder whose diameter is 6 cm and height 21 cm.



31. Given that $\pi = \frac{22}{7}$, then the volume of the cylinder, in cm^3, is

(A) 120
(B) 396
(C) 594
(D) 2 376

32. In a rectangular garden plot, 15 m long and 12 m wide, an area of 80 m^2 is used for a vegetable garden. What area of the plot is NOT used for vegetable gardening?

(A) 26 m^2
(B) 100 m^2
(C) 134 m^2
(D) 260 m^2

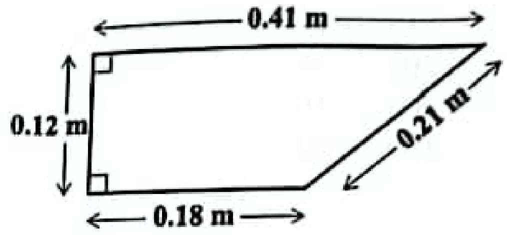
33. How long will a speedboat take to travel between two harbours which are 1 080 km apart, if it travels at an average speed of 120 kmh^{-1}?

(A) 8 hours
(B) 9 hours
(C) 12 hours
(D) 15 hours

34. The lengths of the sides of a triangle are x, $2x$ and $2x$ centimetres. If the perimeter is 20 centimetres, what is the value of x?

(A) 4
(B) 5
(C) 8
(D) 10

Item 35 refers to the following diagram of a trapezium.

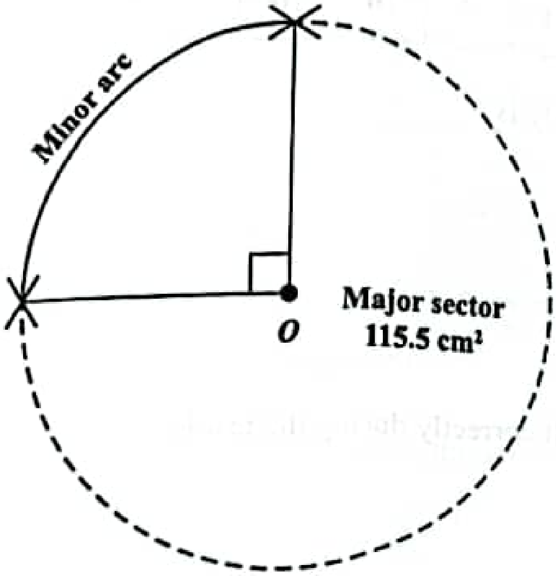


35. The perimeter of the trapezium, in millimetres, is

(A) 0.092
(B) 9.2
(C) 92
(D) 920

Item 36 refers to the following diagram which shows a circle, centred at O. The major sector and minor arc are indicated.

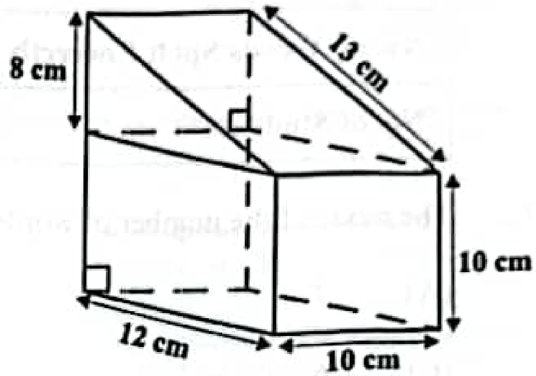
(Use $\pi = \frac{22}{7}$).



36. If the area of the major sector is 115.5 cm^2, then the **diameter** of the circle, in cm, is

(A) 5.5
(B) 7.0
(C) 14.0
(D) 28.0

Item 37 refers to the following diagram which shows a compound solid.



37. The surface area of the compound solid, in cm^2, is

(A) 746
(B) 866
(C) 980
(D) 1 100

38. The distance around a lake is 8 km. On a map, this distance around the lake is represented by a length of 4 cm. The scale on the map is

(A) 1 : 40
(B) 1 : 2 000
(C) 1 : 100 000
(D) 1 : 200 000

Items 39 and 40 refer to the following table which shows the number of words that a group of 60 students got correct in a spelling test that consisted of 10 words.

No. of Words Spelt Correctly	6	7	8	9	10
No. of Students	3	18	7	16	16

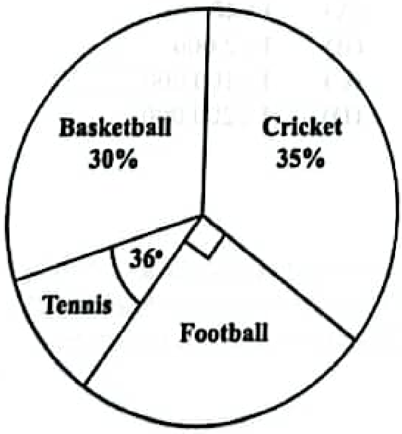
39. The mode of the number of words spelt correctly is

(A) 7
(B) 8
(C) 16
(D) 18

40. The median number of words the students spelt correctly during the test is

(A) 8
(B) 9
(C) 16
(D) 18

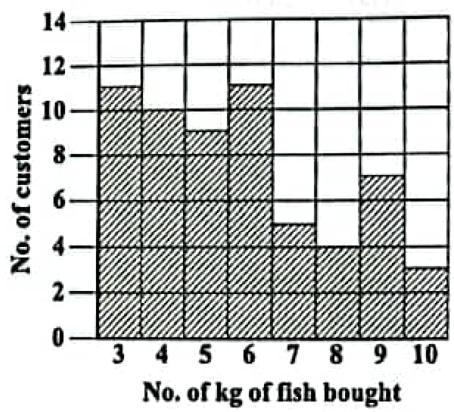
Item 41 refers to the following pie chart which shows the popular games played by a group of students.



41. If 180 students play football, how many students are in the group?

(A) 300
(B) 360
(C) 720
(D) 900

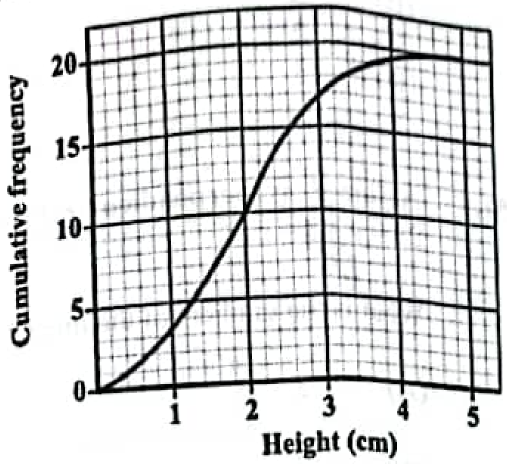
Item 42 refers to the following chart which shows the amount of fish bought, in kg, by the first 60 customers at a fish market.



42. How many customers bought at LEAST 6 kg of fish?

(A) 18
(B) 19
(C) 30
(D) 34

Item 43 refers to the following diagram which shows the cumulative frequency curve of the heights of 20 seedlings.



43. The semi-interquartile range of the set of measurements is

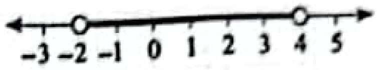
(A) 0.65 cm
(B) 1.30 cm
(C) 2.00 cm
(D) 2.60 cm

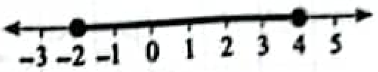
44. Six hundred students sit an examination. The probability of a randomly selected student failing the examination is $\frac{1}{5}$.

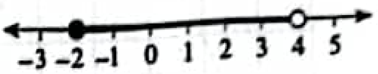
How many students are expected to pass the examination?

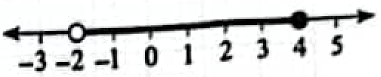
(A) 100
(B) 120
(C) 480
(D) 500

45. Which of the following line graphs represents $\{x: -2 < x \leq 4\}$?

(A) 

(B) 

(C) 

(D) 

46. Which of the following represents the equation of a straight line?

(A) $y = \frac{4}{x}$

(B) $y = 2x + 3$

(C) $y = x^2 - 4$

(D) $y = x^2 + 2x - 5$

47. What is the gradient of the straight line $2y = -3x - 8$?

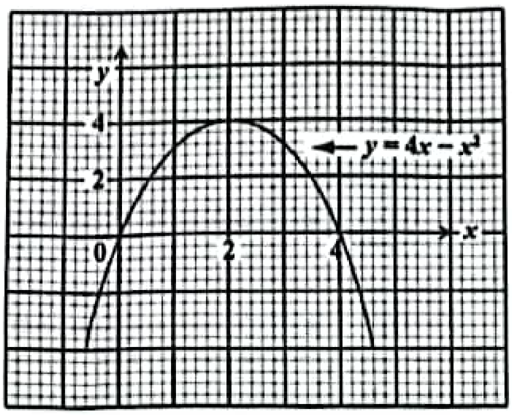
(A) 2

(B) 3

(C) -3

(D) $-\frac{3}{2}$

Items 48 and 49 refer to the following graph of a quadratic function.



48. The maximum point of $y = 4x - x^2$ is

(A) (0, 0)
(B) (0, 4)
(C) (4, 2)
(D) (2, 4)

49. According to the graph, the solution of the equations $y = 4x - x^2$ and $y = 0$ are

(A) $x = 0$ when $y = 0$ and $x = 4$ when $y = 0$

(B) $x = 0$ when $y = 0$ and $x = 2$ when $y = 4$

(C) $x = 2$ when $y = 4$ and $x = 4$ when $y = 0$

(D) $x = 0$ when $y = 0$ and $x = -4$ when $y = -32$

50. If $h(x) = 4x^2 - 6$, then $h\left(-\frac{1}{2}\right) =$

(A) -7
(B) -5
(C) 2
(D) 5

51. A line L is perpendicular to the line $2x - y - 8 = 0$.

What is the gradient of the line L?

(A) -2

(B) $-\frac{1}{2}$

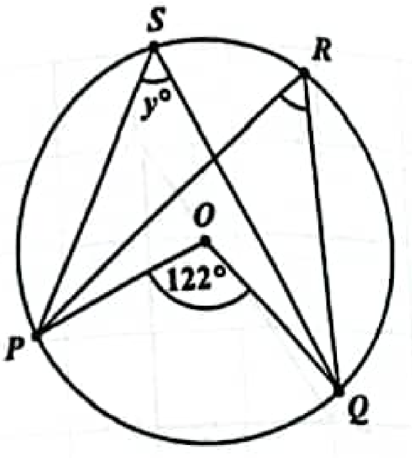
(C) $\frac{1}{2}$

(D) 2

52. Which of the following sets is represented by the function $f: x \to x^2 + 3$ where $x \in \{0, 1, 2, 3\}$?

(A) $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$
(B) $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
(C) $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
(D) $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$

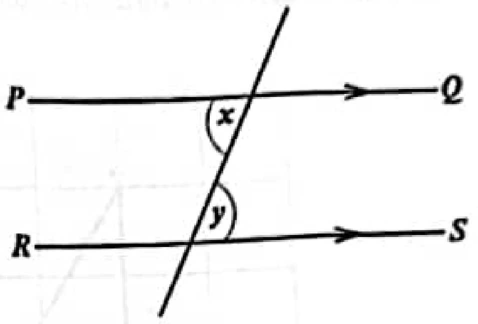
Item 53 refers to the following diagram of a circle with its centre at O. Angle $POQ = 122°$.



53. The value of $y°$ is

(A) 29°
(B) 58°
(C) 61°
(D) 84°

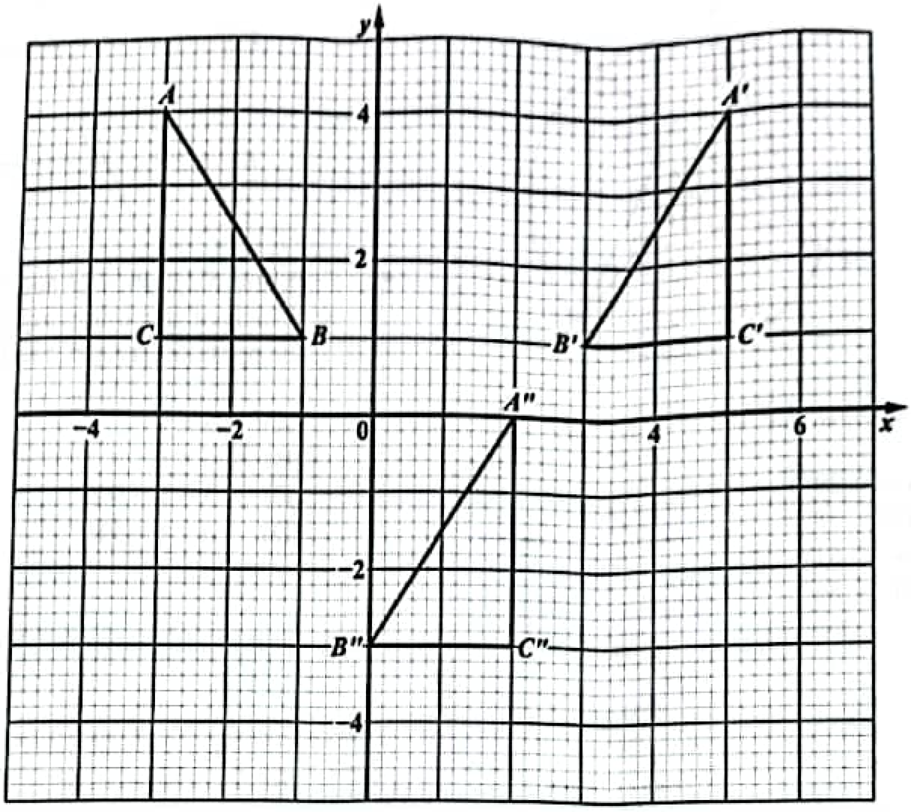
Item 54 refers to the following diagram.



54. In the diagram, PQ and RS are parallel. Which of the following BEST describes the relation between x and y?

(A) $x = y$
(B) $x < y$
(C) $x + y = 90°$
(D) $x + y = 180°$

Item 55 refers to the following diagram which shows Triangle ABC and its images $A'B'C'$ and $A''B''C''$ after Triangle ABC undergoes a composite/double transformation.



55. What sequence of transformations will map triangle ABC onto its image, triangle $A''B''C''$?

(A) A reflection in the line $x = 1$, followed by a translation of $\begin{pmatrix} -3 \\ -4 \end{pmatrix}$

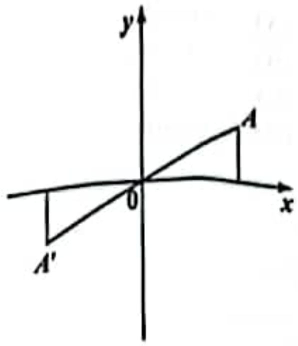
(B) A translation of $\begin{pmatrix} -3 \\ -4 \end{pmatrix}$ followed by a reflection in the x-axis

(C) A translation of 4 units to the left, followed by a clockwise rotation of 90° about the origin

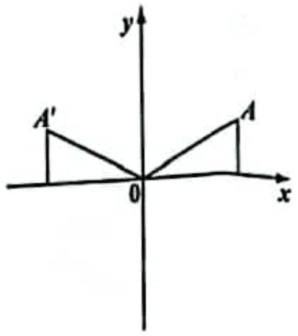
(D) A counterclockwise rotation of 90° about the origin followed by a translation of 4 units downwards

In each of the following diagrams, A' is the image of A. Which of the diagrams shows a reflection in the x-axis?

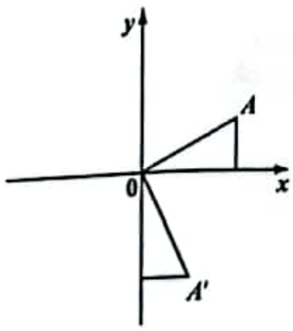
(A)



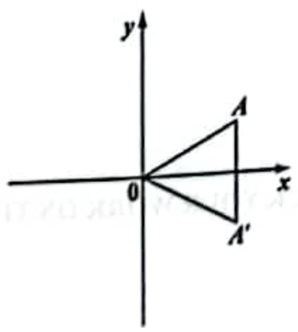
(B)



(C)



(D)



57. In Triangle ABC, Angle $A = x°$ and Angle $B = 2x°$. What is the size of Angle C?

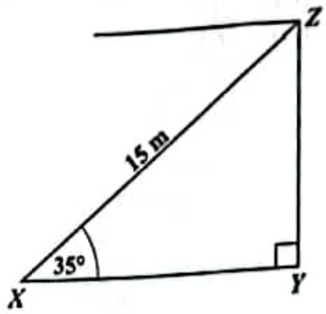
(A) $(180 - 3x)°$

(B) $60°$

(C) $30°$

(D) $\left(\frac{180}{3x}\right)°$

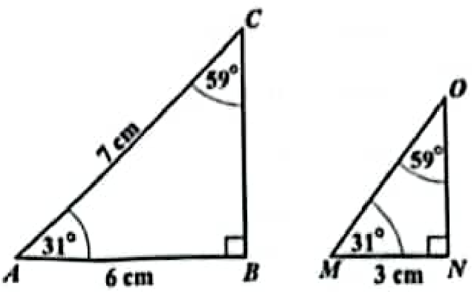
Item 58 refers to the following diagram which shows the angle of elevation of a point, Z, from X.



58. The angle of elevation of the point Z from X is 35°. If X is 15 metres from Z, then the height YZ, in metres, is

(A) 15 cos 35°
(B) 15 sin 55°
(C) 15 sin 35°
(D) 15 tan 55°

Item 59 refers to the following pair of similar triangles.



59. The length of *MO*, in centimetres, is

(A) 3.0
(B) 3.5
(C) 4.6
(D) 6.0

60. Under the translation $\begin{pmatrix}-2\\3\end{pmatrix}$, the image of $(-5, 3)$ is

(A) $(0, -3)$
(B) $(1, -2)$
(C) $(-7, 6)$
(D) $(3, 6)$

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.